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School of Computing and Mathematical Sciences

CO7201 Individual Project

Preliminary Report

**CareMate+: A Secure Android Healthcare Companion for
Medication Management, Remote Monitoring, Pharmacy
Fulfilment and Care Coordination**

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Executive Technical Snapshot

Project at a Glance

CareMate+ is a secure role-based Android healthcare companion designed as a software development project with technical evaluation. The system supports five user groups: patient, doctor, caregiver, pharmacy staff and administrator. The project brings together several healthcare-support workflows that are often handled separately, including medication management, OCR-assisted medicine capture, reminder tracking, simulated vitals monitoring, safety escalation, doctor consultation, pharmacy order tracking and professional account verification.

The application will be developed using React Native for the Android frontend, a Node.js/Express REST API for the backend, PostgreSQL for persistent storage and Prisma as the database ORM. Security is a core design concern. The system will use password hashing, JWT authentication, role-based access control, ownership checks, verification-status restrictions and audit logging to protect sensitive workflows.

MVP Boundary

The MVP will demonstrate the core patient journey from medicine entry to reminders, simulated vitals and safety escalation. It will also include doctor review/consultation, caregiver visibility for linked patients, pharmacy order status tracking and admin verification. Health Connect, external medicine APIs, cloud deployment, analytics and PDF export will remain recommended or optional extensions.

Safety Boundary

CareMate+ is a healthcare-support prototype, not a diagnostic, prescribing or emergency-service replacement. OCR is assistive only and must be reviewed before saving. MVP vitals will be simulated so normal, warning and critical safety workflows can be tested safely without real medical-device data.

CareMate+ Project Snapshot		
Secure role-based Android healthcare companion for medication management, monitoring, consultation and care coordination.		
PLATFORM React Native Android mobile application	BACKEND Node.js / Express REST API	DATABASE PostgreSQL with Prisma ORM
CORE ROLES Patient, Doctor, Caregiver, Pharmacy Staff, Admin	KEY WORKFLOWS Medication management, OCR confirmation, reminders, simulated vitals, Safety Response, consultation, pharmacy tracking	SECURITY BOUNDARY JWT authentication, bcrypt hashing, RBAC, ownership checks, verification gating, audit logs
MVP BOUNDARY: Prototype support tool only — no diagnosis, no emergency-service replacement, no automatic prescribing.		

1. Aims and Objectives

1.1 Project Motivation

Patients taking regular medicines must manage medicine names, strengths, dosages, timings, reminders and follow-up actions. This becomes harder when doctors, caregivers, pharmacies and administrators are involved, because care activities can become fragmented across separate systems.

Medication non-adherence affects around 50% of patients with chronic conditions and contributes to poorer health outcomes [1]. CareMate+ addresses this through a secure role-based Android healthcare companion that combines medication management, OCR-assisted capture, reminders, simulated vitals, Safety Response escalation, doctor consultation, caregiver visibility and pharmacy order tracking.

CareMate+ will not diagnose conditions, prescribe independently or replace emergency services. It is a healthcare-support prototype with clear role responsibilities and access boundaries.

1.2 Project Aim

The aim of CareMate+ is to design, implement and evaluate a secure role-based Android healthcare companion that supports medication management, OCR-assisted medicine capture, reminder tracking, simulated vitals monitoring, Safety Response escalation, doctor consultation, pharmacy fulfilment and care coordination.

1.3 Project Objectives

1. Develop secure registration and login for patient, doctor, caregiver, pharmacy and admin users.
2. Implement JWT-protected APIs with role-based access control, ownership checks and verification-status restrictions.
3. Build patient medicine management with add, view, edit and delete functionality.
4. Implement medication reminders with taken, missed and snoozed dose tracking.
5. Add OCR-assisted medicine or prescription scanning with editable confirmation before saving.
6. Implement simulated vitals for normal, warning and critical readings.
7. Build the Safety Response workflow with countdown, cancellation and escalation.
8. Support doctor workflows for assigned patient review, prescriptions, notes, alerts and consultations.
9. Support caregiver and pharmacy workflows for linked-patient monitoring and order tracking.
10. Implement admin verification and audit logging for sensitive actions.

1.4 User Personas and Stakeholders

Table 1.1: User Personas and Access Boundaries — CareMate+ Role-Based Access Model			
ROLE	MAIN NEED	ACCESS BOUNDARY	KEY SCREENS / FEATURES
 Patient	Manage medicines, reminders, vitals, consultations and pharmacy orders	Own records only	Dashboard, Medicines, Scanner, Reminders, Vitals, Safety Response, Consultations, Pharmacy Orders
 Doctor	Review assigned patients, alerts, notes and prescriptions	Assigned patients only	Doctor Dashboard, Patient Overview, Safety Alerts, Prescriptions, Consultation, Notes
 Caregiver	Monitor linked patient reminders, missed doses and safety status	Linked patients only	Caregiver Dashboard, Linked Patient Summary, Reminders, Vitals Summary, Alerts, Orders
 Pharmacy Staff	Process prescription or refill orders	Relevant pharmacy orders only	Pharmacy Dashboard, Order Details, Accept / Reject Order, Status Update, Order History
 Admin / Verifier	Verify professional accounts and manage access	Verification and account management only	Admin Dashboard, Doctor Verification, Pharmacy Verification, User Status, Audit Logs
NOTE: Access is restricted by role, ownership, relationship and verification checks.			

Table 1: User Personas and Access Boundaries

1.5 End-to-End Care Journey

The CareMate+ journey starts when a patient registers and logs in. The patient can add medicines manually or scan a medicine/prescription image. OCR output appears as an editable draft, so details must be checked before saving. Once confirmed, the system creates reminder schedules and records dose outcomes such as taken, missed or snoozed.

The patient can also view simulated vitals. Normal readings are stored, while critical readings trigger the Safety Response countdown. If cancelled, the event is recorded without escalation. If not cancelled, the system creates a safety alert for the assigned doctor and linked caregiver. The doctor can review the alert, start a consultation, add notes or create a prescription update. Pharmacy staff can process related orders and update their status. Key actions are stored in history and audit logs.

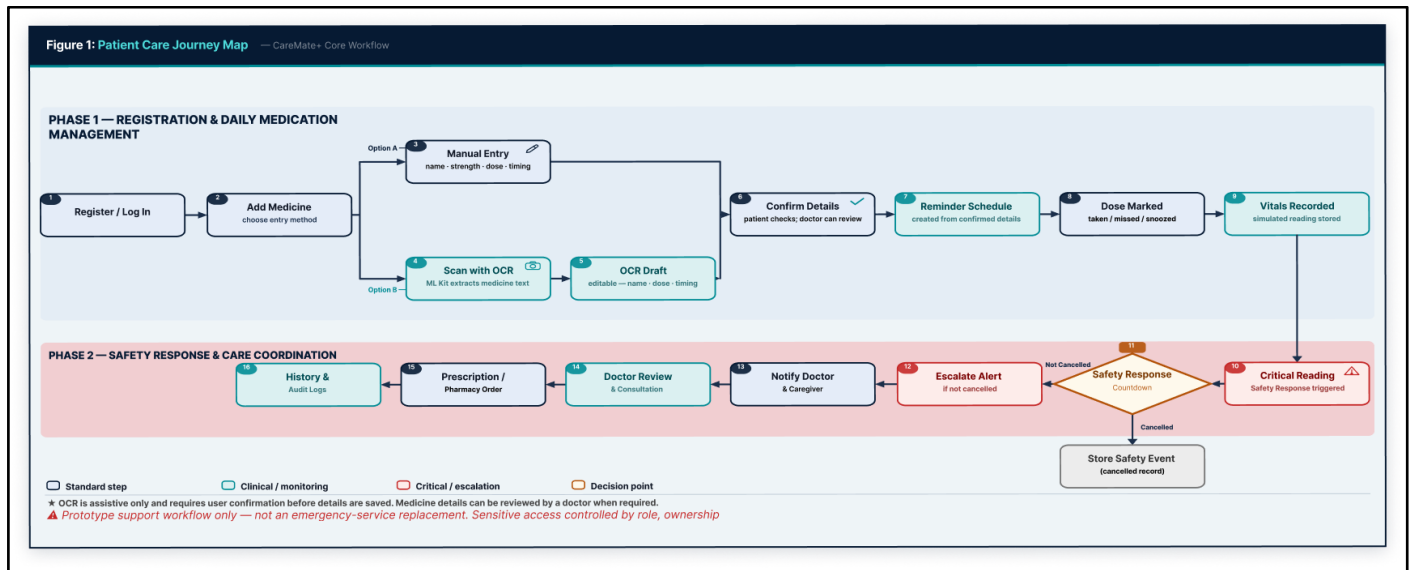


Figure 1: Patient Care Journey Map

1.6 MSc-Level Technical Challenges

1. **Multi-role access control:** Patients, doctors, caregivers, pharmacy staff and admins need different permissions. **Mitigation:** central RBAC middleware.
2. **Ownership and relationship checks:** RBAC alone is insufficient for patient data. **Mitigation:** record ownership, doctor-assignment and caregiver-link checks.
3. **Professional verification gating:** Doctors and pharmacy staff must be approved before sensitive access. **Mitigation:** admin-controlled verification status.
4. **Safe OCR handling:** OCR text may be inaccurate. **Mitigation:** editable draft confirmation before saving.
5. **Safety Response workflow:** Critical simulated vitals require controlled escalation. **Mitigation:** explicit normal, warning, critical, countdown, cancel and escalate states.
6. **Integrated care coordination:** Medicines, reminders, dose logs, vitals, alerts, consultations, prescriptions, orders and audit logs must work together. **Mitigation:** consistent relational schema and audit logging.

2. Requirements

2.1 Requirement Classification

CareMate+ requirements are grouped as essential, recommended and optional. Essential requirements define the MVP. Recommended requirements improve security, usability and coordination after the MVP is stable. Optional requirements are stretch goals attempted only if time remains.

2.2 Essential Requirements

ID	Essential Requirement	Acceptance Evidence
SEC-01	Support registration and login for patient, doctor, caregiver, pharmacy and admin users.	All roles log in.
SEC-02	Hash passwords and protect APIs with JWT authentication.	Invalid tokens rejected.
SEC-03	Enforce role-based access control.	Wrong roles blocked.
SEC-04	Check patient ownership, doctor assignment and caregiver links before returning data.	Unauthorised access denied.
SEC-05	Block doctor/pharmacy workflows until admin approval.	Pending users restricted.
SEC-06	Store data in PostgreSQL using Prisma relationships.	Data persists.
SEC-07	Audit sensitive actions: verification, prescriptions, alerts and order updates.	Audit logs created.
PAT-01	Show patient dashboard with medicines, reminders, vitals, consultations and orders.	Dashboard loads.
PAT-02	Allow patients to add, view, edit and delete medicines.	CRUD works.
PAT-03	Allow OCR scanning of medicine/prescription images.	OCR draft produced.
PAT-04	Allow OCR output to be edited before saving.	Output editable.
PAT-05	Allow reminders and dose states: taken, missed, snoozed.	Dose saved.
PAT-06	Show simulated vitals and trigger Safety Response for critical readings.	Response starts.
DOC-01	Allow verified doctors to access only assigned patients.	Unassigned access denied.
DOC-02	Allow doctors to review assigned patient medicines, reminders, vitals, scans and consultations.	History visible.
DOC-03	Allow doctors to create/upload prescription details.	Prescription saved.
DOC-04	Allow doctors to review prescription/OCR details before medicine/reminder updates.	Draft produced.
DOC-05	Allow doctors to manage consultations, notes, alerts and video links.	Workflow completes.
CAR-01	Allow caregivers to view linked-patient medicines, reminders, missed doses, vitals and alerts.	Linked data visible.
CAR-02	Allow caregivers to view consultation and pharmacy order status.	Status displayed.
PHA-01	Allow pharmacy staff to view prescription, refill and medicine orders.	Orders displayed.

ID	Essential Requirement	Acceptance Evidence
PHA-02	Allow pharmacy staff to accept, reject and update order status.	Status updates.
ADM-01	Allow admins to review doctor/pharmacy verification details, documents and register links.	Request displayed.
ADM-02	Allow admins to approve or reject professional accounts.	Status changes.

Table 2 : Essential Requirements Summary

2.3 Recommended Requirements

ID	Recommended Requirement	Acceptance Evidence
REC-01	Support email verification.	Flow works.
REC-02	Support password reset with time-limited tokens.	Expired token rejected.
REC-03	Save confirmed OCR details as medicine/reminder records.	Records created.
REC-04	Let patients share scans with assigned doctors.	Scan visible to doctor.
REC-05	Show patient adherence history.	History shown.
REC-06	Let patients request refills/medicine orders.	Request created.
REC-07	Let patients add health notes/care diary entries.	Note saved.
REC-08	Show doctors assigned patient summary reports.	Summary generated.
REC-09	Let doctors add advice/notes to scanned medicine information.	Note saved.
REC-10	Let doctors request follow-up actions/appointments.	Follow-up recorded.
REC-11	Let caregivers add linked-patient care notes.	Note saved.
REC-12	Let caregivers request linked-patient consultations.	Request created.
REC-13	Let pharmacy staff add order notes, rejection reasons and collection instructions.	Notes visible.
REC-14	Show admin dashboard summaries for verifications, professionals and activity.	Summary displayed.

Table 3 : Recommended Requirements Summary

2.4 Optional Requirements

ID	Optional Requirement	Acceptance Evidence
OPT-01	Integrate Health Connect for supported device readings.	Data read.
OPT-02	Integrate an external medicine API.	Data retrieved.
OPT-03	Add doctor adherence and missed-dose analytics.	Chart shown.
OPT-04	Add vitals and safety-alert trend views.	Trend shown.
OPT-05	Add adherence, missed-dose, alert and vitals heatmaps.	Heatmap shown.
OPT-06	Add simple text/log analysis for notes or safety events.	Summary produced.
OPT-07	Deploy backend, database and API to cloud.	Endpoint works.

Table 4: Optional Requirements Summary

2.5 Requirement Rationale

Essential requirements prove the core CareMate+ MVP: secure multi-role access, medicine capture, reminders, simulated vitals, Safety Response, doctor review, caregiver awareness, pharmacy fulfilment and admin verification. Recommended requirements improve security, usability and coordination. Optional requirements are separated because they depend on time, integrations or deployment.

Requirement IDs make testing clearer because each feature can be checked against acceptance evidence without mixing MVP work with stretch goals.

Advanced MVP Feature	Feature Purpose	Linked Requirements	Why It Is Different	Why It Is Technically Difficult	Level
Patient medicine/ prescription scanning	Allows a patient to scan a medicine label or prescription image so medicine information can be drafted instead of typed fully manually.	PAT-03, PAT-04	Uses OCR to convert an image into editable medicine information.	OCR output can be inaccurate, so the system must create an editable draft, allow correction and only save confirmed data.	High
Doctor prescription scanning/ review	Allows a verified doctor to upload or review prescription information for an assigned patient.	DOC-03, DOC-04	Connects doctor-side prescription handling with patient medicine and reminder workflows.	It requires verified doctor access, assigned-patient checks, prescription storage and safe confirmation before updates are applied.	High
Prescription-to-reminder update	Converts confirmed prescription details into patient reminder information.	DOC-03, DOC-04, PAT-05	Links doctor-created prescription details directly to the patient reminder workflow.	The system must map medicine name, dosage and timing into reminder fields without automatically trusting OCR or prescription text.	High
Safety Response workflow	Responds to simulated critical vitals using countdown, cancellation and escalation behaviour.	PAT-06, DOC-05, CAR-01	Goes beyond storing vitals by creating a controlled safety workflow.	It needs state-based logic for normal, warning, critical, countdown, cancel and escalate outcomes.	High
Video consultation support	Allows a verified doctor and assigned patient to start or access a consultation link from	DOC-05	Adds communication support inside the care workflow instead of keeping	It requires secure consultation access, correct doctor-patient linking and	Medium

	inside the CareMate+ workflow.		video calls separate from patient records and alerts.	fallback handling if Jitsi/JaaS setup is delayed.	
Multi-role care coordination	Allows patient, doctor, caregiver, pharmacy and admin workflows to operate together with different permissions.	DOC-01, DOC-02, CAR-01, CAR-02, PHA-01, PHA-02, ADM-01, ADM-02	Connects several user roles instead of building a single-user medicine reminder app.	It requires RBAC, ownership checks, doctor assignment, caregiver linking, pharmacy status updates and admin verification to work together correctly.	High

Table 5: Technically Advanced MVP Features

3. Technical Specification

3.1 System Overview

CareMate+ uses a client-server architecture. A React Native Android app connects to a Node.js/Express REST API. PostgreSQL stores users, medicines, reminders, vitals, alerts, prescriptions, pharmacy orders and audit logs, while Prisma manages schema and queries.

3.2 Technology Stack Overview

Area	Technology	Purpose
Mobile App	React Native	Android role-based screens.
Backend API	Node.js and Express	REST APIs and workflow logic.
Database	PostgreSQL	Relational healthcare-support data.
ORM	Prisma	Schema, migrations and queries.
Authentication	JWT and bcrypt	Protected routes and password hashing.
Access Control	RBAC and ownership checks	Role, assignment and linked-user restrictions.
OCR	Google ML Kit OCR	Medicine/prescription text extraction.
Video	Jitsi/JaaS	Doctor-patient video consultation.
Testing	Postman and manual testing	API and scenario validation.
Version Control	GitLab	Code, commits and evidence.

Table 6: Technology Stack Overview

3.3 Target System Architecture

The Android app sends requests to the Express API, which checks authentication, permissions and ownership before Prisma reads or writes PostgreSQL data. OCR and Jitsi/JaaS support scanning and video consultation.

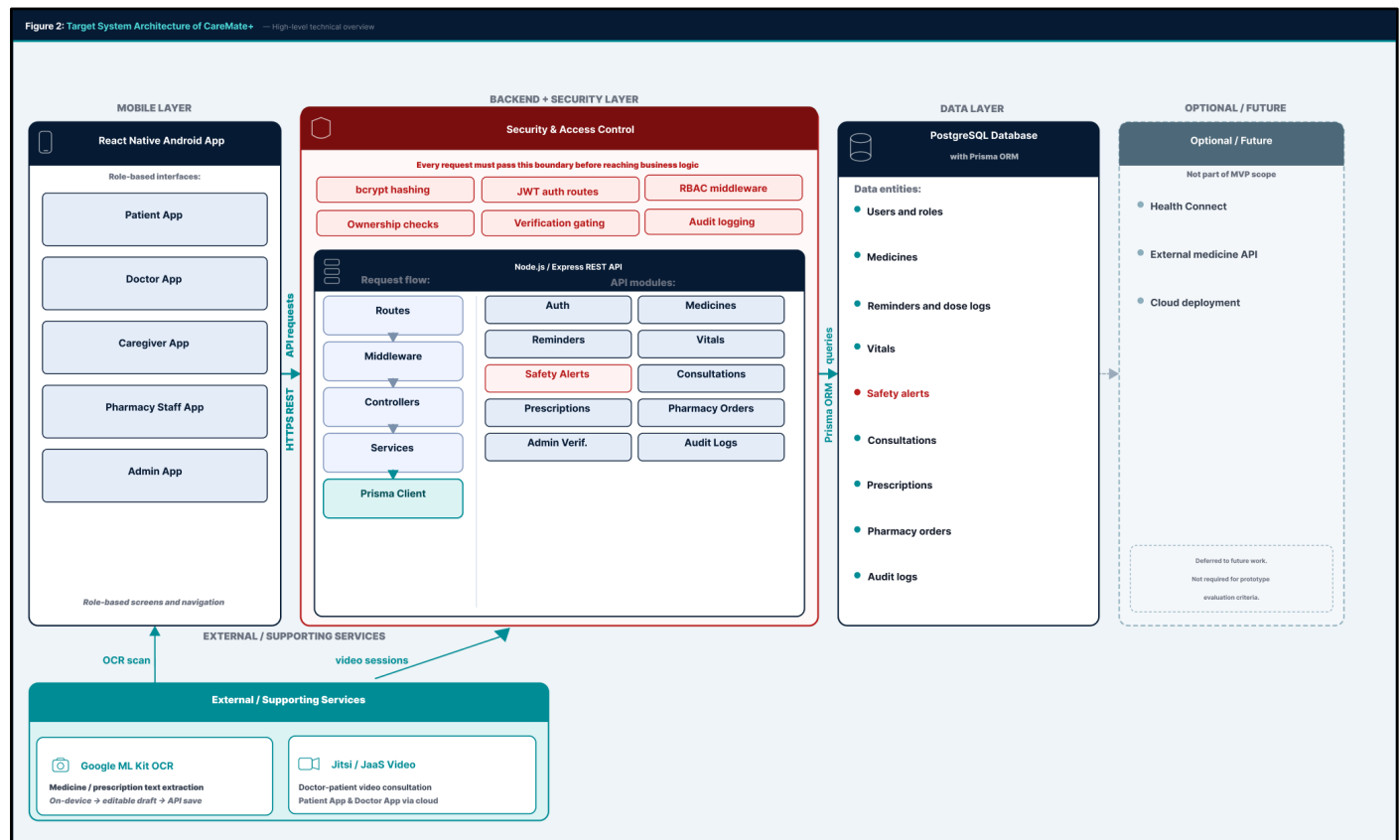


Figure 2 : Target System Architecture of CareMate+.

3.4 Security and Access Control Model

CareMate+ secures patient-related data using JWT authentication, bcrypt password hashing, role-based access control, ownership/assignment checks, admin verification gating and audit logging for sensitive actions.

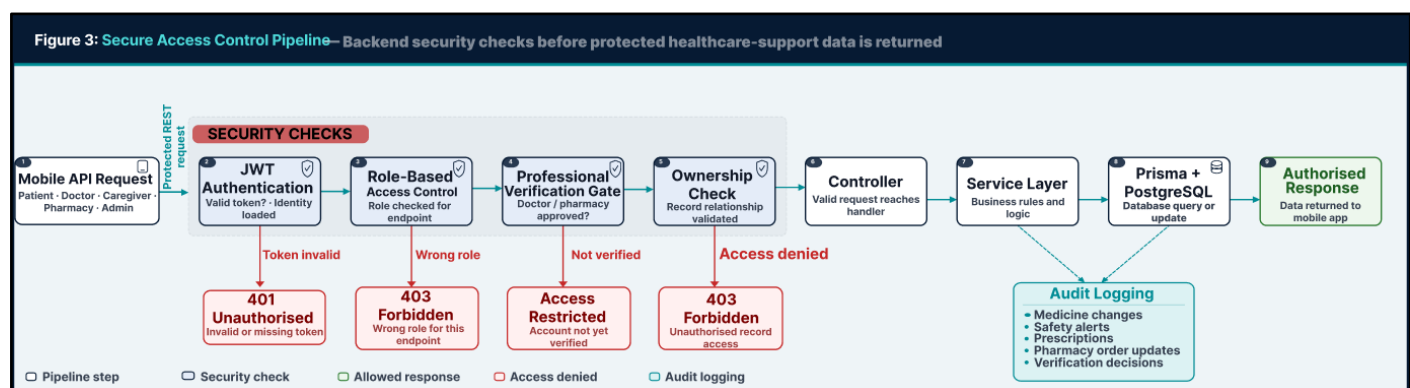


Figure 3 : Secure Access Control Pipeline

3.5 Main Technical Workflows

The main workflows are shown below to explain OCR-based reminder creation, doctor prescription-to-reminder updates and Safety Response escalation.

3.5.1 OCR-to-Reminder Workflow

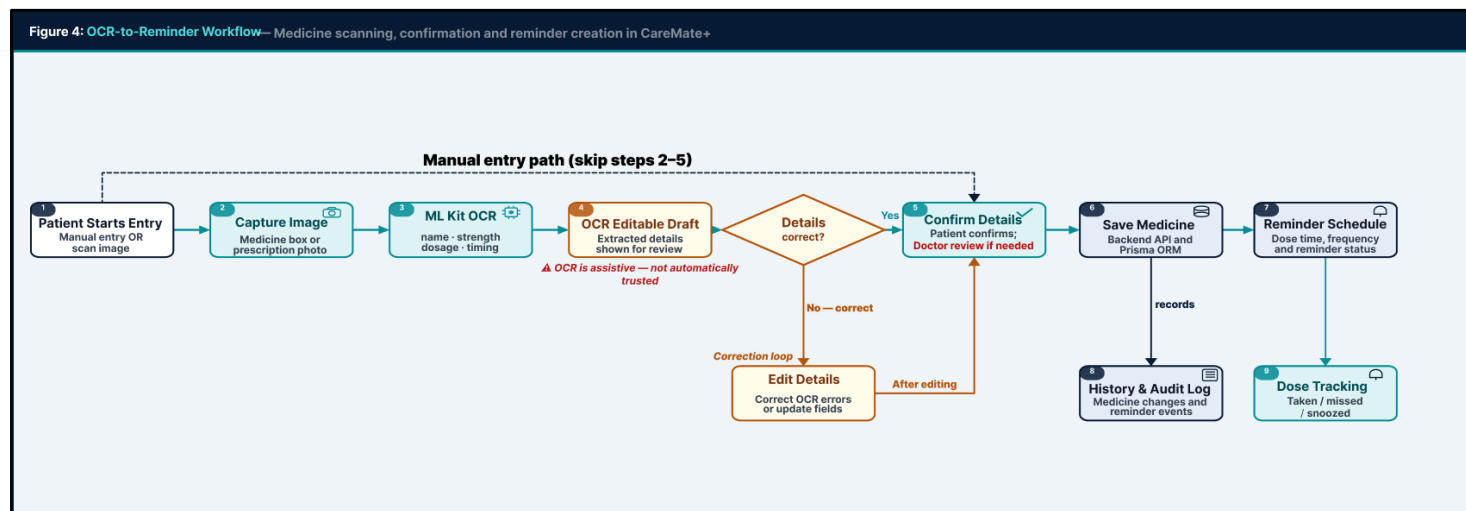


Figure 4: OCR-to-Reminder Workflow

3.5.2 Doctor Prescription-to-Reminder Workflow

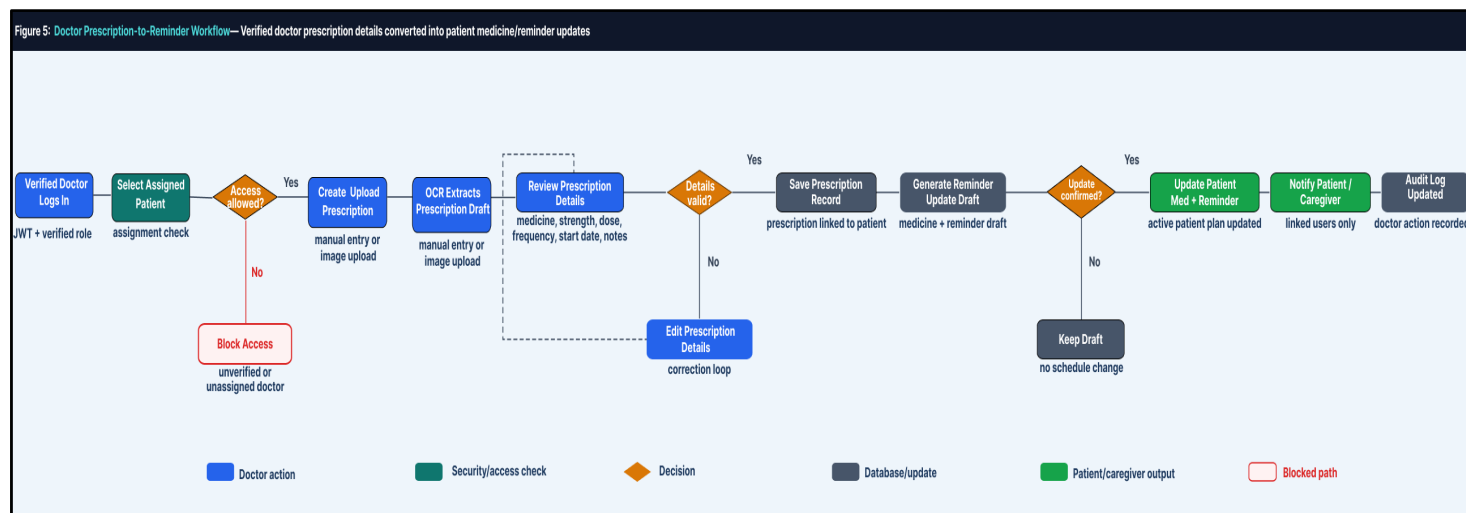


Figure 5 : Doctor Prescription-to-Reminder Workflow

3.5.3 Safety Response State Machine

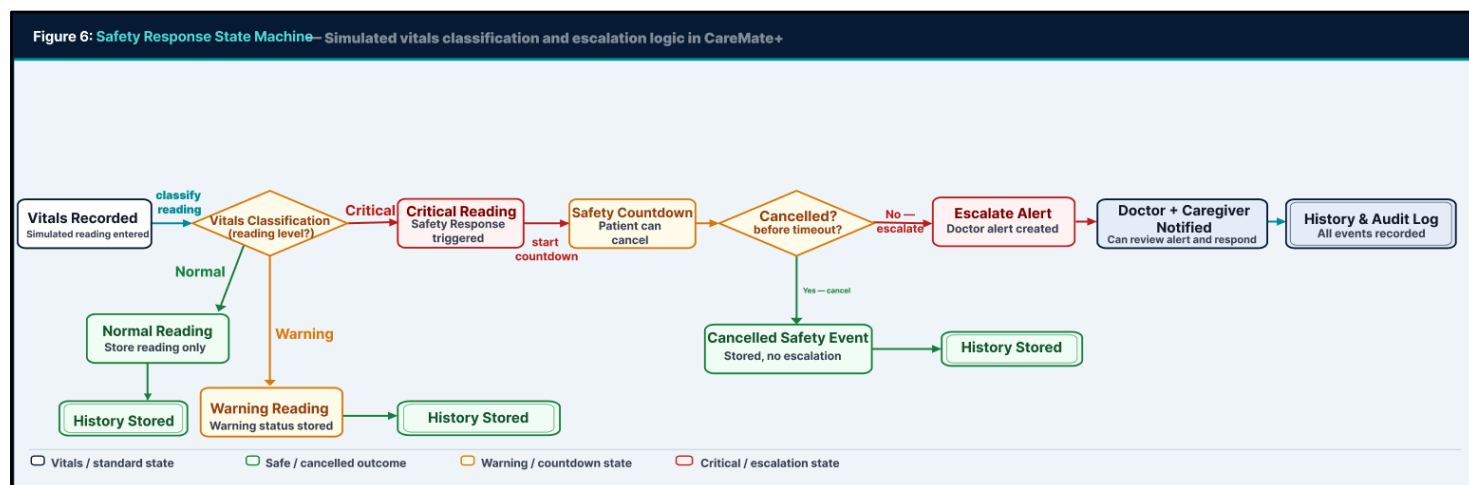


Figure 6 : Safety Response State Machine

4. Requirements Evaluation Plan

This section explains how CareMate+ will be evaluated against the requirements defined earlier. The evaluation will check whether the prototype works correctly for each role, protects patient-related records, supports the core medicine and safety workflows, and can be demonstrated through repeatable test scenarios.

4.1 Evaluation Strategy

CareMate+ will be evaluated using functional testing, security testing, scenario-based workflow testing, OCR testing, Safety Response timing testing and usability evaluation. The focus will be on the essential requirements because they define the minimum successful prototype. Technical testing will use the Android application for user-facing workflows, Postman for backend API checks, and Prisma Studio or PostgreSQL queries for database verification. Participant-based usability testing will be conducted after ethics approval is granted.

Testing evidence will be recorded through pass/fail logs, mobile screenshots, API responses, database checks, observation notes and questionnaire results where relevant. Progress and results will be reviewed during supervisor meetings where appropriate.

4.2 Evaluation Criteria Matrix

Evaluation Area	Requirement Link	Method	Measurable Success Criteria	Evidence
Authentication and protected routes	SEC-01, SEC-02	Login and API testing	Protected routes reject missing or invalid JWT tokens.	API responses and pass/fail log.
Role-based access control	SEC-03, SEC-05	Negative access testing	Wrong-role access attempts are blocked.	API responses and screen evidence.
Ownership and relationship checks	SEC-04	Security scenario testing	Patient-to-patient, doctor-to-unassigned-patient and caregiver-to-unlinked-patient access is blocked.	API responses, database checks and pass/fail log.
Medicine and reminder management	PAT-01, PAT-02, PAT-05	Functional testing	Patient can manage medicines, create reminders and record taken, missed or snoozed doses.	Screenshots, API responses and database records.
OCR medicine/prescription capture	PAT-03, PAT-04, DOC-04	OCR scenario testing	Clear sample images produce useful OCR drafts, and OCR output remains editable before saving.	OCR test log, draft screenshots and corrected records.
Safety Response workflow	PAT-06, DOC-05, CAR-01	Simulated vitals testing	Critical readings start countdown, cancellation prevents escalation and expiry creates authorised alerts.	Timer screenshots, alert records and pass/fail log.
Doctor prescription-to-reminder workflow	DOC-01, DOC-03, DOC-04	End-to-end scenario testing	Verified doctor creates/uploads prescription details for assigned patient and generates a reminder update draft.	Scenario checklist, screenshots and database records.
Multi-role care coordination	DOC-02, CAR-02, PHA-01, PHA-02, ADM-01, ADM-02	End-to-end scenario testing	Doctor, caregiver, pharmacy and admin users complete at least one authorised workflow each.	Screen/API evidence and scenario checklist.
Participant usability testing	User-facing workflows	Task-based testing and SUS questionnaire after ethics approval	Participants complete representative tasks using demo accounts and simulated data; SUS responses and observation notes are recorded.	Observation notes, SUS responses and task completion log.
Usability fallback review	User-facing workflows	Nielsen heuristic review if ethics approval is delayed	Interface is reviewed for navigation clarity, readable labels, status feedback, OCR confirmation and Safety Response clarity.	Heuristic checklist and improvement log.

Table 7 : Evaluation Criteria Matrix

4.3 Functional Testing

Functional testing will check whether the main application features work as expected. This will include registration, login, medicine creation, reminder creation, dose tracking, simulated vitals entry, consultation requests, pharmacy order updates and admin verification actions. Each test will compare the expected result with the actual mobile screen, backend response and database record.

A feature will be considered successful if the action is completed, the backend returns the correct response and the related database record is created or updated correctly.

4.4 Security Testing

Security testing will focus on authentication, authorisation and access boundaries. JWT-protected routes will be tested to confirm that unauthenticated users cannot access protected endpoints. Role-based access control will be tested by attempting to access features using the wrong role. Ownership and relationship checks will test cases such as a patient accessing another patient's medicines, a doctor viewing an unassigned patient and a caregiver viewing an unlinked patient.

The expected result is that invalid or unauthorised attempts are blocked with an appropriate response, such as 401 Unauthorised or 403 Forbidden.

4.5 Scenario Testing

Scenario testing will evaluate complete user journeys rather than isolated screens. Example scenarios include a patient scanning a medicine and creating a reminder, a critical vital reading triggering a Safety Response alert, a doctor reviewing an assigned patient, a caregiver viewing a linked-patient alert, a pharmacy updating an order and an admin approving or rejecting a professional account. Another scenario will test a doctor creating or uploading prescription details for an assigned patient and checking whether the related medicine and reminder information is created or updated correctly.

Each scenario will be successful if the correct role can complete the workflow, unauthorised roles are blocked and the expected database records or status updates are produced.

4.6 OCR Evaluation

The OCR feature will be evaluated using a small test set of clear medicine or prescription images. The target is that at least 85% of samples extract useful medicine-related text, such as medicine name, strength, dosage or timing. CareMate+ will not automatically trust OCR output, so 100% of OCR results must appear as editable drafts before saving.

The OCR workflow will therefore be successful when users can review, correct and confirm extracted information before a medicine record or reminder is created.

4.7 Safety Response Timing Evaluation

The Safety Response workflow will be evaluated using simulated normal, warning and critical vitals. Normal and warning readings should be stored without escalation. Critical readings should start a countdown within 2 seconds of classification. If the patient cancels within the countdown, no alert should be escalated. If the countdown expires, the system should create an alert for the assigned doctor and notify the linked caregiver.

The workflow will be successful if the countdown, cancellation and escalation behaviour match the Safety Response state machine.

4.8 Usability Evaluation

CareMate+ will aim to include participant-based usability testing after ethics approval. The planned method is task-based usability testing with a short SUS questionnaire. Participants will use demo accounts and simulated data to complete representative role-based tasks, such as adding a medicine, confirming OCR output, checking reminder status, reviewing a safety alert, updating a pharmacy order or approving a professional account.

The ethics application will be prepared before any participant testing begins. This will include the online ethics form, participant information sheet, consent form, study protocol and draft usability tasks/questionnaire. No participant data will be collected before ethics approval is granted. If ethics approval is delayed, usability evaluation will be limited to Nielsen's heuristic review and developer-led walkthroughs.

5. Background Research and Reading List

This section explains the background research behind CareMate+. It links the project to existing healthcare-support problems, relevant technologies and limitations in current systems. The section does not claim clinical effectiveness. It only justifies why the prototype includes reminder support, OCR confirmation, controlled access, simulated monitoring, consultation support and pharmacy tracking. The sources guide implementation choices and later testing evidence.

5.1 Medication Management and Adherence

Medication adherence is a common difficulty for patients managing long-term conditions, multiple medicines and repeated dosage schedules. Missed or incorrect doses can reduce treatment effectiveness and make self-management harder [1]. A systematic review and meta-analysis also found that mobile app interventions can improve medication adherence, although evidence quality and context still matter [2]. CareMate+ addresses this by supporting medicine records, reminder scheduling and dose-status tracking for taken, missed and snoozed doses.

5.2 OCR for Medicine and Prescription Capture

Manual medicine entry can be slow and may introduce typing errors. OCR can support medicine capture by extracting text from medicine packaging or prescription images. Google ML Kit Text Recognition provides image-based text extraction for mobile applications [3]. However, OCR can be inaccurate, so CareMate+ treats extracted text as an editable draft rather than trusted data. Users must confirm or correct the draft before saving medicines or reminders. Academic OCR research shows that recognition accuracy can vary depending on input quality, so CareMate+ uses editable OCR drafts rather than automatically saving extracted medicine or prescription text [19].

5.3 Healthcare Security and Role-Based Access

Healthcare-support systems must protect patient-related records. OWASP guidance highlights the need for strong authorisation controls [4]. CareMate+ therefore combines JWT-protected routes, role-based access control and ownership checks. Patients access their own records, doctors access assigned patients, caregivers access linked patients, and pharmacy/admin users are restricted to their workflows. Research on mobile health

apps has identified privacy and security risks, which supports the need for authentication, role checks, ownership checks and audit logging in CareMate+ [20].

5.4 Remote Monitoring and Vitals

Remote monitoring can support early awareness of patient changes through readings such as heart rate, blood pressure, oxygen saturation, glucose and temperature. Android Health Connect shows how mobile platforms can support health data through permission-controlled APIs [5]. For feasibility, CareMate+ uses simulated vitals in the MVP. This still allows the prototype to demonstrate normal, warning and critical readings and the Safety Response workflow.

5.5 Teleconsultation and Care Coordination

Care often involves communication between patients, doctors, caregivers and pharmacies. Jitsi/JaaS can support embedded video consultations without building a full video infrastructure [6]. In CareMate+, video consultation is a supporting feature connected to doctor review, alerts, medicines and patient history. Telemedicine research suggests that remote consultation can support healthcare delivery, but its effectiveness depends on context and implementation quality [21].

5.6 Pharmacy Fulfilment Workflow

Medication support also includes prescription or refill progress. NHS Electronic Prescription Service guidance shows the importance of electronic prescribing and pharmacy fulfilment workflows [7]. CareMate+ uses a simplified version where pharmacy staff can accept, reject and update order status as pending, accepted, preparing, ready or completed. This also supports the CareMate+ prescription-to-reminder workflow, where prescription details created or uploaded by a doctor can be linked to patient medicine schedules and pharmacy order progress.

5.7 Existing System Comparison

System Type / Example Systems	Typical Focus	Limitation	CareMate+ Response
Medication reminder apps: Medisafe [13], MyTherapy [14]	Reminders and medication tracking	Mainly focused on patient-side medication support, not a full patient-doctor-caregiver-pharmacy-admin workflow.	Adds OCR, doctor review, caregiver visibility and pharmacy order tracking.
OCR tools: Google Lens [15], Google ML Kit OCR [3]	Text extraction from images	Not directly linked to medicine records, reminders or care workflows.	Uses editable OCR drafts before saving medicine or prescription information.
Teleconsultation platforms: Jitsi/JaaS [6]	Video calls	Separate from reminders, vitals and patient medication records.	Links consultation with patient records, alerts and assigned doctor workflows.
Pharmacy systems: NHS Electronic Prescription Service [7]	Prescription and order fulfilment	Focuses on prescription transmission and pharmacy workflow, with limited monitoring context.	Shows order progress within wider care coordination.

System Type / Example Systems	Typical Focus	Limitation	CareMate+ Response
Digital healthcare apps: NHS App [16]	Access to official healthcare services	Official service access, but not designed as a custom prototype for OCR, simulated vitals, Safety Response and multi-role testing.	Combines medicines, OCR, reminders, vitals, alerts, consultation, pharmacy tracking and multi-role workflows.

Table 8: Existing System Comparison

5.8 Reading List

The following sources will guide the implementation and evaluation of CareMate+:

- **WHO medication adherence report** — used to justify the medication reminder and adherence-support features [1].
- **Peng et al. systematic review on mobile adherence apps** — used as academic evidence that mobile applications can support medication adherence [2].
- **Google ML Kit Text Recognition documentation** — used to support OCR-assisted medicine and prescription capture [3].
- **OWASP authorisation guidance** — used to support JWT-protected routes, RBAC and ownership checks [4].
- **Android Health Connect documentation** — used as background for health data and future vitals integration [5].
- **Jitsi/JaaS documentation** — used to support doctor-patient video consultation [6].
- **NHS Electronic Prescription Service guidance** — used to understand digital prescription and pharmacy fulfilment workflows [7].
- **React Native, Express, Prisma and PostgreSQL documentation** — used to support the mobile app, backend API and database design [8]-[12].
- **Medisafe and MyTherapy** — used to compare CareMate+ with existing medication reminder and adherence-support applications [13], [14].
- **Google Lens** — used to compare general image/text recognition tools with CareMate+'s editable OCR confirmation workflow [15].
- **NHS App** — used to compare CareMate+ with an official digital healthcare access application [16].
- **Nielsen's usability heuristics and SUS** — used to support the fallback usability review and short usability questionnaire after ethics approval [17], [18].
- **Additional academic papers** on OCR, mobile health security and telemedicine are used to strengthen the technical background for OCR validation, secure role-based workflows and video consultation [19]-[21].

6. Time-plan and Risk Plan

This section explains how CareMate+ will be managed. The project follows an incremental, risk-first approach: backend, database, authentication and access control will be completed before OCR, Safety Response, role-based screens, pharmacy fulfilment and admin verification. Optional features will only be attempted after the essential MVP is stable.

6.1 Project Management Approach

Weekly targets, Git commits, supervision checkpoints and testing evidence will manage progress. Each week should show an API, mobile screen, database update, diagram, test result or report section. Initial backend/authentication work has started alongside the preliminary report.

Supervision will review scope, progress and MVP risk. Screens will be built in order: patient, OCR/vitals, Safety Response, doctor prescription, caregiver, pharmacy and admin.

6.2 Ethics Approval and Participant Testing Plan

Participant-based usability testing will only be conducted after ethics approval. The application will include the participant information sheet, consent form, study protocol and usability task/SUS questionnaire materials. Testing will use demo accounts and simulated data only. No real patient data, prescriptions or medical records will be used. Participants will complete representative tasks such as adding a medicine, confirming OCR output, checking reminders, reviewing a safety alert, updating a pharmacy order or approving a professional account.

If ethics approval is delayed, participant testing will not be conducted. Usability will instead be evaluated through Nielsen's heuristic review and developer-led walkthroughs.

6.3 12-Week Project Roadmap

Implementation, evidence collection and writing-up will run in parallel throughout the project. Each week will produce development progress and report material, such as screenshots, API responses, database checks, diagrams, testing notes, supervisor feedback, ethics/usability materials and evaluation writing. The target is a feature-complete MVP by Week 10, with Weeks 11-12 reserved for testing, final report polish and viva/demo preparation.

Week	Date range	Implementation focus	Report / evidence
1	15-21 Jun	Scope, repository, backend setup, database planning and patient-screen planning.	Draft aims, objectives, project scope and initial requirements.
2	22-28 Jun	Authentication, email verification, basic patient screens and admin doctor/pharmacy verification start.	Complete preliminary report, diagrams, requirement tables and risk plan for 24 Jun supervision.
3	29 Jun-5 Jul	Authentication, email verification, Prisma/PostgreSQL, admin approve/reject and patient dashboard/medicine screens.	Update technical specification, architecture evidence and early screenshot/API evidence. Start ethics preparation.
4	6-12 Jul	RBAC, ownership checks, verification gating, reminders and doctor assigned-patient screens.	8 Jul supervision: security evidence, admin verification evidence, early UI evidence and updated implementation notes.

5	13-19 Jul	Medicine CRUD, reminders, dose states, OCR confirmation and doctor patient-review screens.	Patient/doctor workflow evidence, evaluation-plan update and report writing notes for implemented features.
6	20-26 Jul	Vitals, Safety Response countdown/cancel, doctor alert review and consultation screens.	22 Jul supervision: OCR, vitals, Safety Response and doctor workflow evidence. Draft usability task/SUS materials.
7	27 Jul-2 Aug	Most patient/doctor MVP workflows, Safety Response escalation, prescription upload and early prescription-to-reminder update.	Principal marker checkpoint: 50-60% MVP evidence, workflow screenshots, API evidence and progress write-up.
8	3-9 Aug	Complete prescription-to-reminder, scanned prescription review and caregiver linked-patient screens.	5 Aug supervision: doctor/caregiver review, caregiver evidence, ethics approval status check and updated evaluation notes.
9	10-16 Aug	Pharmacy order screens, admin dashboard improvement, audit logging and integration fixes.	Pharmacy/admin evidence, audit-log evidence, interim/progress report update and evaluation evidence collection.
10	17-23 Aug	Caregiver reminders/alerts, pharmacy statuses, admin refinements, audit logs, Jitsi links and final API-screen integration.	19 Aug supervision: feature-complete MVP review, integration evidence, usability setup and draft final-report structure.
11	24-30 Aug	Functional, security, scenario and database testing; screenshot/API evidence collection.	Draft evaluation chapter, test-result summaries, evidence appendix and usability testing or fallback review evidence.
12	31 Aug-6 Sep	Final bug fixes, report polish, Git cleanup, references, demo script and viva preparation; optional features if time permits.	2 Sep supervision: final review, final report polish, Git evidence, demo script and viva material.

Table 9 : 12-Week Project Roadmap

Figure 7: 12-Week Project Roadmap / Gantt Chart
CareMate+ — Secure Android Healthcare Companion

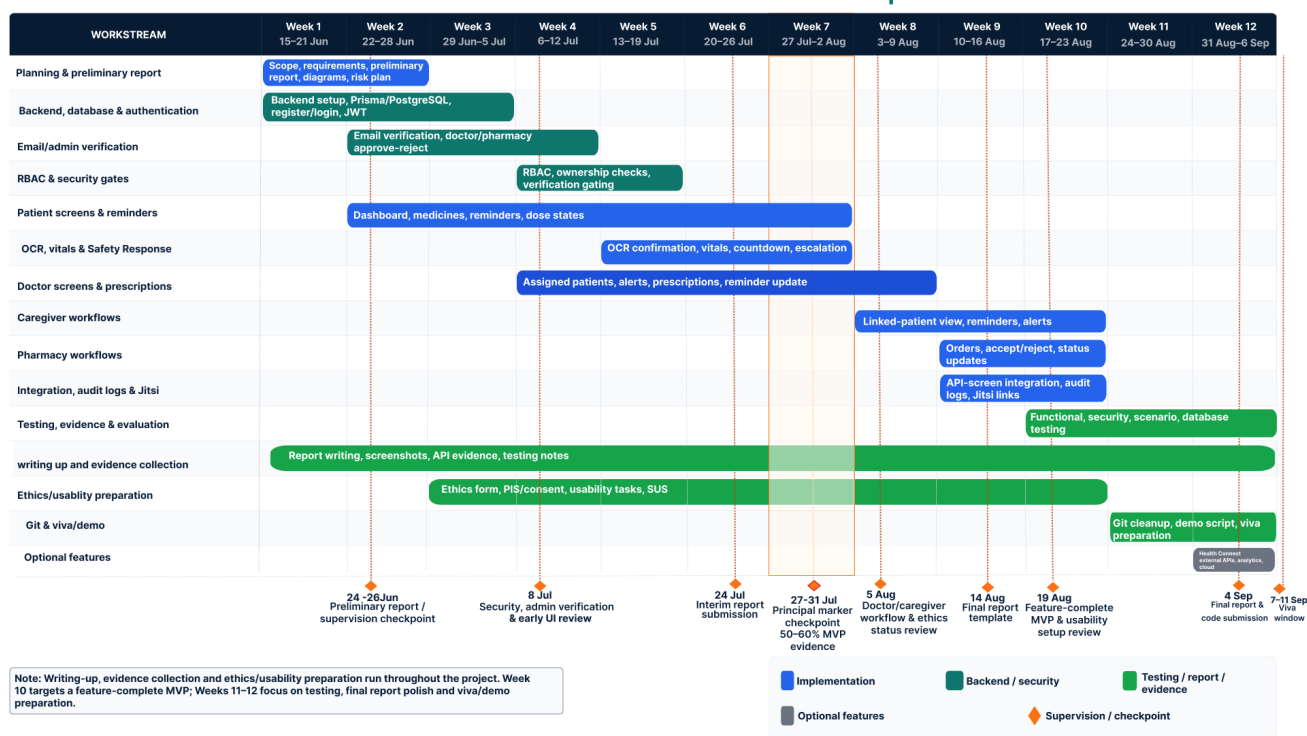


Figure 7 : Project Gantt Chart

6.4 Milestone Gates

Milestone	Target	Success criteria
Scope/repository	Week 1-2	Aims, requirements and initial commits ready.
Preliminary report	Week 2	Report sections, diagrams, evaluation and risk plan prepared.
Authentication/admin verification	Week 2-3	Register/login, email verification, hashing, JWT and admin approve/reject working.
Database/RBAC	Week 3-4	Prisma/PostgreSQL, RBAC, ownership checks and verification gating tested.
Patient medicine/reminders	Week 4-5	Dashboard, medicines, reminders and dose states working.
OCR/Safety Response	Week 5-7	OCR draft, vitals classification and countdown/escalation working.
Patient/doctor checkpoint	Week 7	Most patient and doctor MVP workflows ready for principal marker evidence.
Ethics and usability preparation	Weeks 3-10	Ethics documents, demo-account task materials, SUS questionnaire and usability setup prepared before participant testing.
Integrated multi-role MVP	Week 10	Patient, doctor, caregiver, pharmacy and admin workflows integrated for testing.
Testing/report/viva	Week 11-12	Evidence, final report, Git repository, demo script and viva material ready.

Table 10 : Milestone Gates

6.5 Risk Management Approach

Risk management protects the essential MVP. If time is limited, optional work is reduced first. The MVP must include authentication, email/admin verification, RBAC, medicines, reminders, OCR confirmation, prescription-to-reminder update, simulated vitals, Safety Response and multi-role workflows. Each supervision meeting will review risks and decide whether optional features should be frozen.

6.6 Risk Register with Contingency

Risk	Likelihood	Impact	Mitigation	Contingency
Scope too large	High	High	Prioritise essentials and freeze optional work early.	Remove Health Connect, analytics, cloud and external APIs.
RBAC/ownership errors	Medium	High	Test wrong-role access and ownership checks in Postman.	Fix security before optional work.
Email verification delay	Medium	Medium	Test token flow early with registration/login.	Document limitation and protect login flow.
OCR inaccurate	Medium	Medium	Use editable OCR drafts and confirmation.	Keep manual entry fallback.
Safety workflow complex	Medium	High	Keep states simple: normal, warning, critical, countdown, cancel, escalate.	Use simulated data only.
Prescription workflow too broad	Medium	Medium	Limit to upload/create, review and reminder update.	Remove advanced prescribing features.
Role screens delayed	Medium	Medium	Build screens by priority: patient, doctor, caregiver, pharmacy, admin.	Keep screens simple and finish essentials first.
Jitsi/JaaS delay	Medium	Medium	Add video consultation after core workflows.	Use meeting link fallback.
Ethics approval delay	Medium	Medium	Prepare ethics documents early and review them with the supervisor.	Use Nielsen's heuristic review and developer-led walkthroughs if approval is delayed.
Testing time limited	Medium	High	Collect screenshots, API responses and database evidence during development.	Test essential workflows and security first.

Table 11: Risk Register with Contingency

6.7 Risk Heatmap

The heatmap summarises likelihood and impact. Highest priorities are scope overload, testing pressure, RBAC errors and Safety Response complexity.

Priority	Risks
High likelihood / High impact	Scope overload; testing pressure
Medium likelihood / High impact	RBAC errors; Safety Response complexity
Medium likelihood / Medium impact	OCR inaccuracy; email verification delay; prescription scope; role-screen delays; Jitsi/JaaS delay; ethics approval delay
Low priority	Optional features: Health Connect, external medicine APIs, analytics, cloud deployment

Table 12: Risk Heatmap Categories

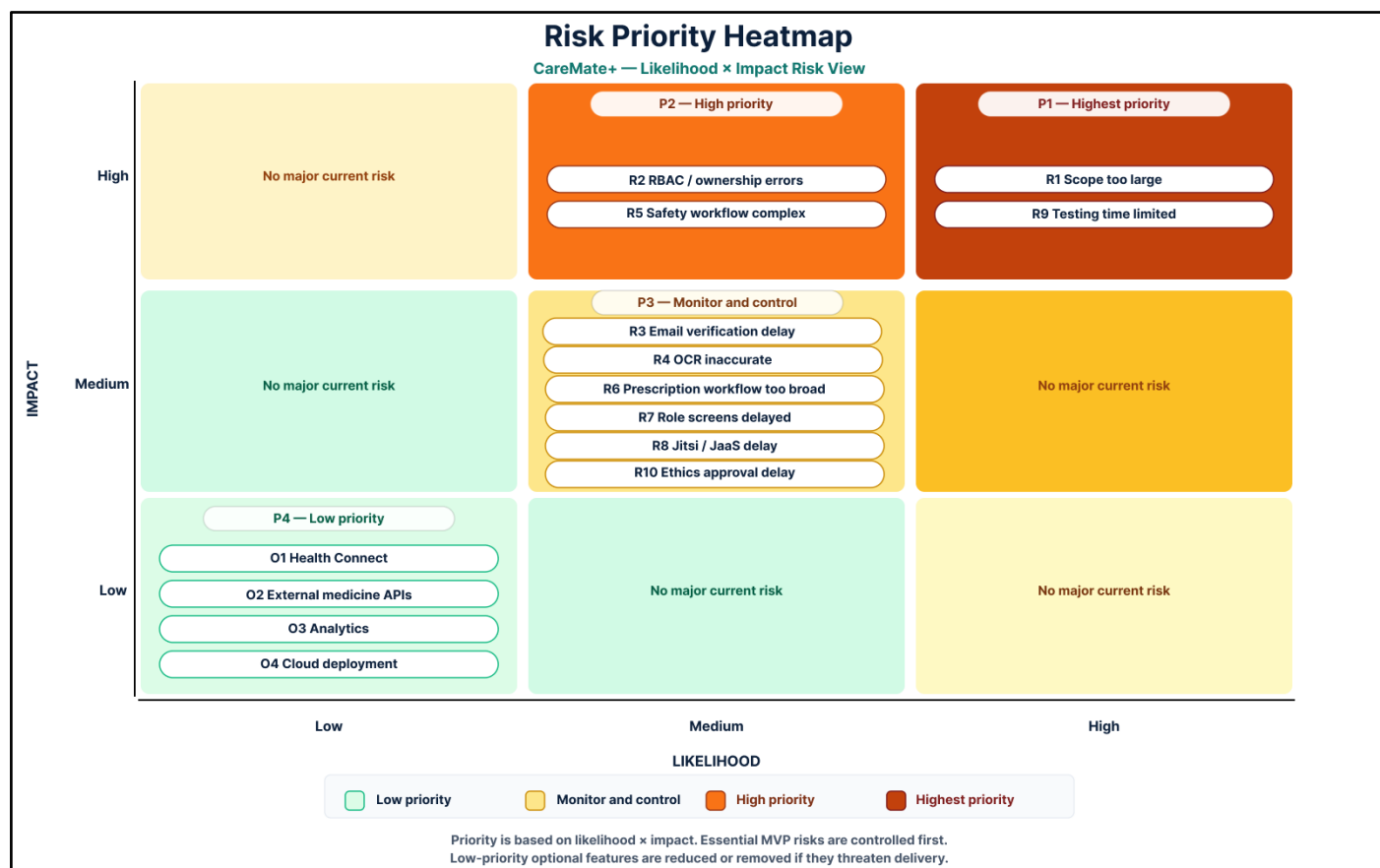


Figure 8 : Risk Priority Heatmap

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